How Dangerous Is Myanmar's Dam Infrastructure?

A Hobby Seismology Project After the 2025 M7.7 Mandalay Earthquake

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https://github.com/akzedevops/mmeqopendata

Catalog through 2026-11-12
Scientific audit: November 2026

Abstract

After the devastating March 2025 earthquake in Myanmar, I wanted to understand how many of the country's 254 dams are actually at risk. This write-up documents a hobby project that pulls earthquake data from a public API, scores every dam for seismic risk, and runs probabilistic hazard analysis—all with open-source Python tools. The Abrahamson & Silva (2008) ground-motion model is compared with a small set of instrumental recordings using event-specific rupture distances and site conditions. The heuristic screening grades classify 63.0% of the inventory as Critical or High; these grades are not probabilities of dam failure. An OpenStreetMap-based scenario analysis estimates PGA above $0.1g$ at 1,674 schools and 583 hospitals. These are modeled exposures, not observed damage. Everything—code, data, interactive maps—is on GitHub.

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1 Background

On March 28, 2025, a magnitude 7.7 earthquake hit central Myanmar along the Sagaing Fault. The rupture was enormous—475 km of surface break, supershear velocity, 5–7 m of slip. The archived Naypyidaw station has two horizontal PGA peaks near $0.57g$ and $0.62g$; their geometric mean is about $0.60g$. The vertical peak is a different component and must not be compared directly with a horizontal GMPE. This paper does not establish disaster casualty totals or observed dam-damage counts.

Myanmar has at least 254 documented dams (irrigation, hydropower, multi-purpose), many built before modern seismic codes existed. The Sagaing Fault runs right through the middle of the country where most of these dams are. I wanted to know: which dams are most at risk, and can we quantify it with open data?

1.1 What This Project Does

- Pulls 10,088 earthquake records (through 2026-11-12) from the Myanmar Earthquake API
- Scores all 254 dams for seismic risk using a proper ground-motion model
- Runs probabilistic hazard curves, Coulomb stress transfer, fragility analysis
- Validates predictions against actual recorded ground motions
- Publishes everything as an open-source toolkit with interactive maps

1.2 Data Sources

The following sources are open inputs or supplementary datasets. The WorldPop raster is not tracked, and the archived finite-fault model is not used by the default Coulomb calculation; neither should be confused with a reproduced input.

Data	Source	License
Earthquakes	Myanmar Earthquake API (USGS/ISC)	Open
Dams (254)	Open Development Mekong (IFC/WLE)	CC BY-SA 4.0
Fault traces (395)	GEM Global Active Faults Database	CC BY-SA 4.0
Site V_{S30}	USGS ShakeMap V_{S30} grid (Wald & Allen 2007)	Public domain
Population grid	WorldPop 2020 (1 km)	CC BY 4.0
ShakeMap stations	USGS ShakeMap (us7000pn9s)	Public domain
Finite fault model	USGS (Goldberg et al. 2025)	Public domain
Surface rupture	USGS (Reitman et al. 2025)	Public domain
Buildings (34,224)	OpenStreetMap (Overpass API)	ODbL

2 The Earthquake Catalog

The catalog has 10,088 events through 2026-11-12. Most are small (mean M 3.21), but there are 6 events $\geq$ M7.0 and 393 $\geq$ M5.0. 73.3% have reported depths below 30 km. Catalog completeness and magnitude scales vary by network and time.

2.1 b-Value and Catalog Completeness

The Gutenberg-Richter law says $\log_{10} N = a - bM$. The tricky part is that the catalog is incomplete at small magnitudes—the 1970s network couldn't detect M2 events, but the modern network can. Fitting the full catalog with automatic completeness gives $b = 0.35$ at $M_c = 2.4$. After Gardner-Knopoff (1974) declustering, $M_c = 4.7$ and $b = 1.04$; these are the rates used for

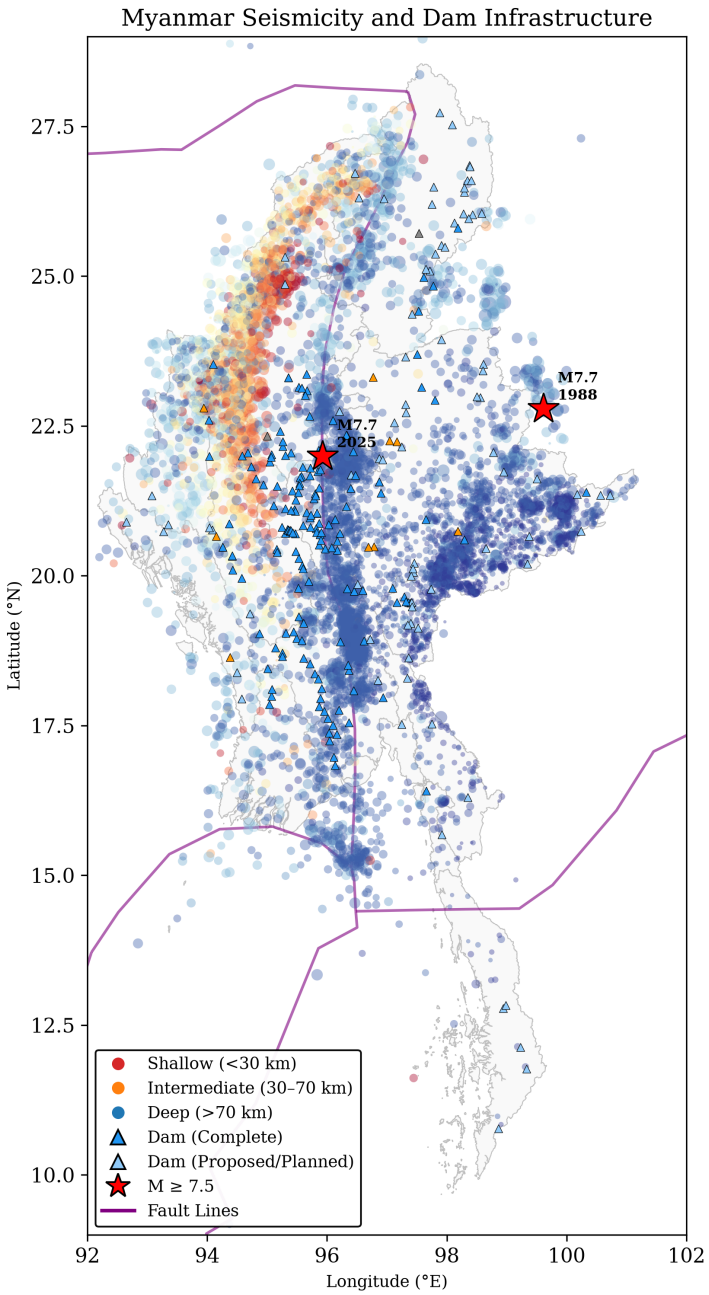


Figure 1: Earthquake distribution, dam locations, and fault lines. The clustering along the Sagaing Fault corridor is obvious.

the probabilistic hazard below. A plausible b-value alone does not establish catalog completeness or a stationary Poisson process.

A stability analysis (Figure 2b) shows the b-value stabilizes around $M_c \approx 4.0$–4.5; the headline fit uses $M_c = 4.0$. Using that cutoff: $b = 0.73$ with 2,805 events. No confidence interval is asserted here without accounting for catalog heterogeneity and magnitude uncertainty.

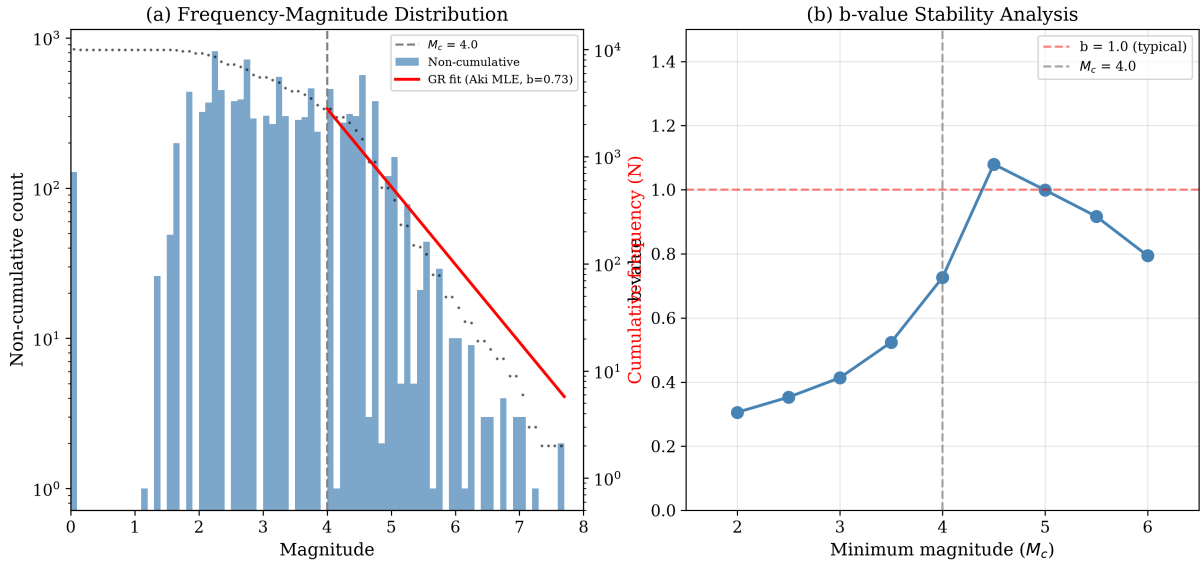


Figure 2: (a) Frequency-magnitude distribution with G-R fit above $M_c = 4.0$. (b) b-value stability—it stabilizes around $M_c \approx 4.0$–4.5, confirming the catalog is incomplete below M4. The G-R line in (a) uses the Aki (1965) MLE b-value at $M_c = 4.0$, the same estimate quoted in the text.

2.2 Spatial Clustering

DBSCAN uses UTM coordinates so distances are metric. The central Myanmar concentration includes the 2025 M7.7 sequence; cluster counts and membership depend on the chosen neighborhood radius and the evolving catalog and should not be interpreted as independent tectonic source zones.

3 Ground Motion: Getting the GMPE Right

This is where I spent the most time. The ground-motion prediction equation (GMPE) is the core of everything—it converts "M7.7 earthquake at X km distance" into "PGA = Yg at this dam."

3.1 Abrahamson & Silva (2008)

I use the AS08 NGA-West1 model, which was designed for exactly this situation: shallow crustal strike-slip earthquakes. The full model has 6 terms:

$$\ln(\text{PGA}) = f_1(M, R_{rup}) + f_{style} + f_{site}(V_{S30}, \text{PGA}_{1100}) + f_{HW} + f_{ZTOR} + f_{large} \quad (1)$$

The coefficients ($a_1 = 0.804$, $a_2 = -0.9679$, $c_1 = 6.75$, $c_4 = 4.5$, $V_{LIN} = 865.1$, $b = -1.186$, etc.) are from Tables 4–5a. An independent computation with OpenQuake 3.26.2 agrees to numerical precision on 80 vertical strike-slip cases spanning M5–8, distances 1–200 km, and $V_{S30} = 180$–1100 m/s, with default basin depth. This checks implementation parity for those

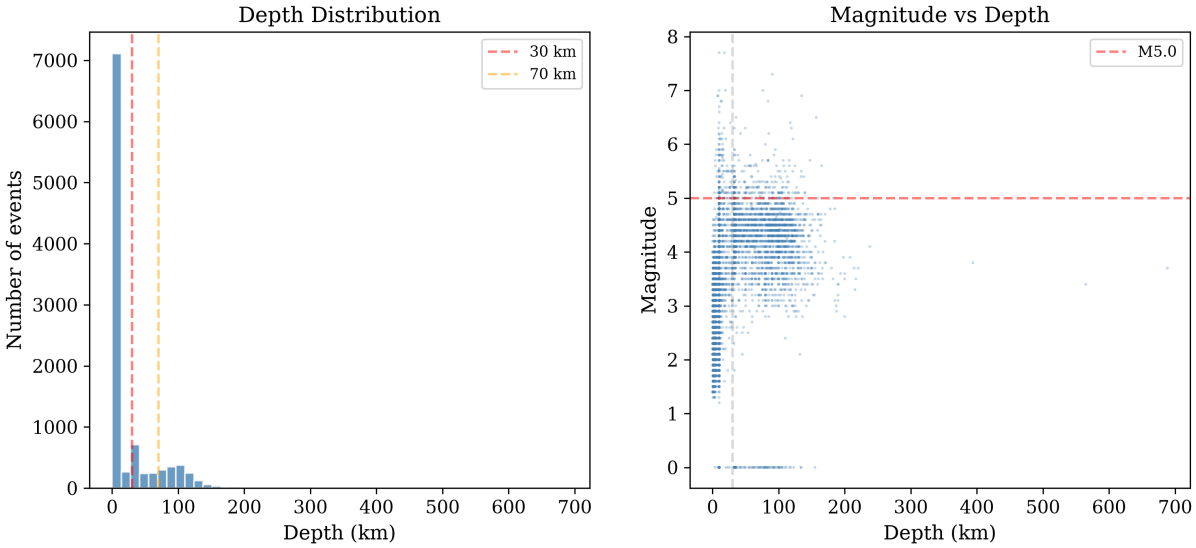


Figure 3: (a) Reported depth distribution. (b) Magnitude versus depth: large events include both shallow and intermediate-depth earthquakes.

cases, not the applicability of AS08 to Myanmar. The local simplified hanging-wall term is not validated for dipping faults.

The nonlinear site response is important—it depends on both V_{S30} and the rock-site PGA, so soft soil sites don't just scale linearly.

3.2 Validation Against Actual Recordings

The archived USGS ShakeMap station list contains six instrumental stations. PANO's horizontal PGA values are flagged as outliers and are excluded, leaving 5 clean horizontal pairs (Table 1). We use the event-specific `distances.rrup`, `distances.rjb`, and station `vs30` supplied by USGS, rather than distance to an unrelated mapped fault. Observed PGA is the geometric mean of the two unflagged horizontal component peaks, not the first channel (which may be vertical). This is an approximation to AS08's rotation-independent GMRotI50; the exact component requires time histories and cannot be reconstructed from scalar peaks alone.

Table 1: AS08 station diagnostic, ordered by USGS rupture distance. Generated from `docs/report/shakemap_validation.csv`. CHTO and KTN exceed the approximate 200 km calibration range and are extrapolations, not in-range validation.

Station	R_{rup} (km)	V_{S30} (m/s)	Obs. (g)	Pred. (g)	Ratio
NPW	2.8	310	0.598	0.478	1.25
NGU	114.0	325	0.049	0.071	0.69
YGN	157.1	263	0.021	0.054	0.40
CHTO	272.2	640	0.010	0.017	0.59
KTN	370.6	400	0.013	0.012	1.07

Only 3 retained stations are within 200 km. This small, single-event sample is a diagnostic, not regional validation of the GMPE, dam fragility, or PSHA. The model is not recalibrated to these observations. An earlier version assigned KTN a 2.3 km distance to an unrelated active fault, despite the supplied event-rupture distance being 370.594 km, and misidentified its vertical channel as the observation. The resulting "50-fold overprediction" and claim that KTN was an interpolated grid value were artifacts of the comparison, not supported scientific conclusions.

Modeled dam PGA is excluded from the observed-data table: comparing one prediction with another is not independent validation. The table is generated directly from the station diagnostic on every publication build.

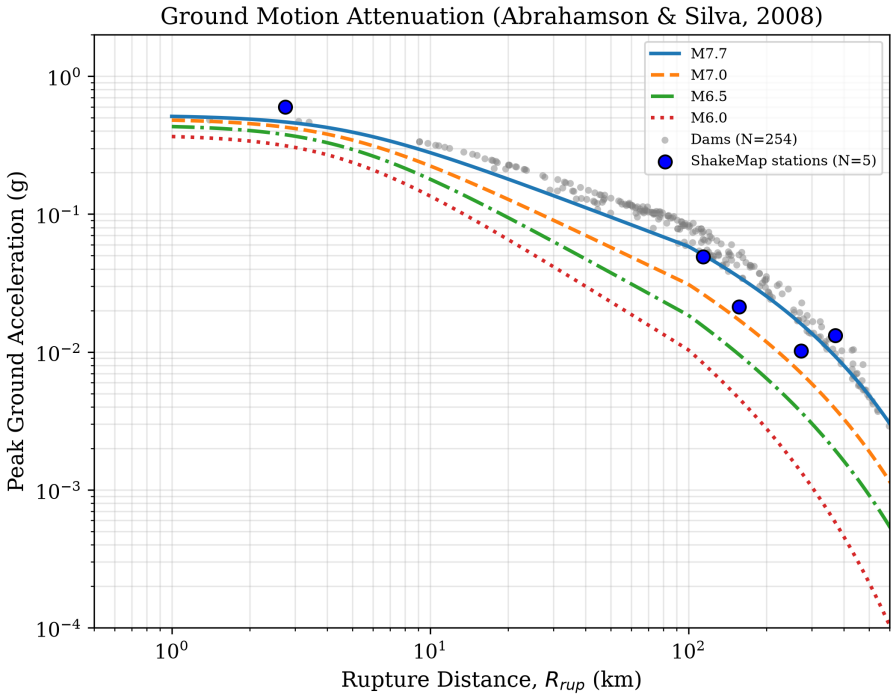


Figure 4: Reference-rock PGA attenuation curves, scenario dam predictions, and unflagged horizontal station observations. Dam and station site conditions vary; the curves use $V_{S30} = 760\,\mathrm{m/s}$.

3.3 Site-Specific Vs30

Instead of assuming all dams sit on rock ($V_{S30} = 760\,\mathrm{m/s}$), I sampled V_{S30} at each site from the USGS ShakeMap V_{S30} grid (a Wald & Allen 2007 topographic-slope proxy). Values range from 229 to 875 m/s (mean 450 m/s). These are site proxies, not measured geotechnical profiles; nonlinear amplification depends on both site velocity and input shaking.

4 Dam Risk Scoring

Each dam gets a composite score (0–10) from four components:

$$R = 0.35 \times S_{seismic} + 0.30 \times S_{fault} + 0.20 \times S_{proximity} + 0.15 \times S_{exposure} \tag{2}$$

Here $S_{seismic}$ is PGA-based. The fault score uses distance to a mapped fault or plate boundary; the proximity score uses distance to the 2025 epicenter. The exposure score combines inventory height, capacity, and storage. The screening traces are not a complete active-fault inventory and are distinct from the actual event rupture used for scenario PGA.

4.1 Results

With the corrected GMPE:

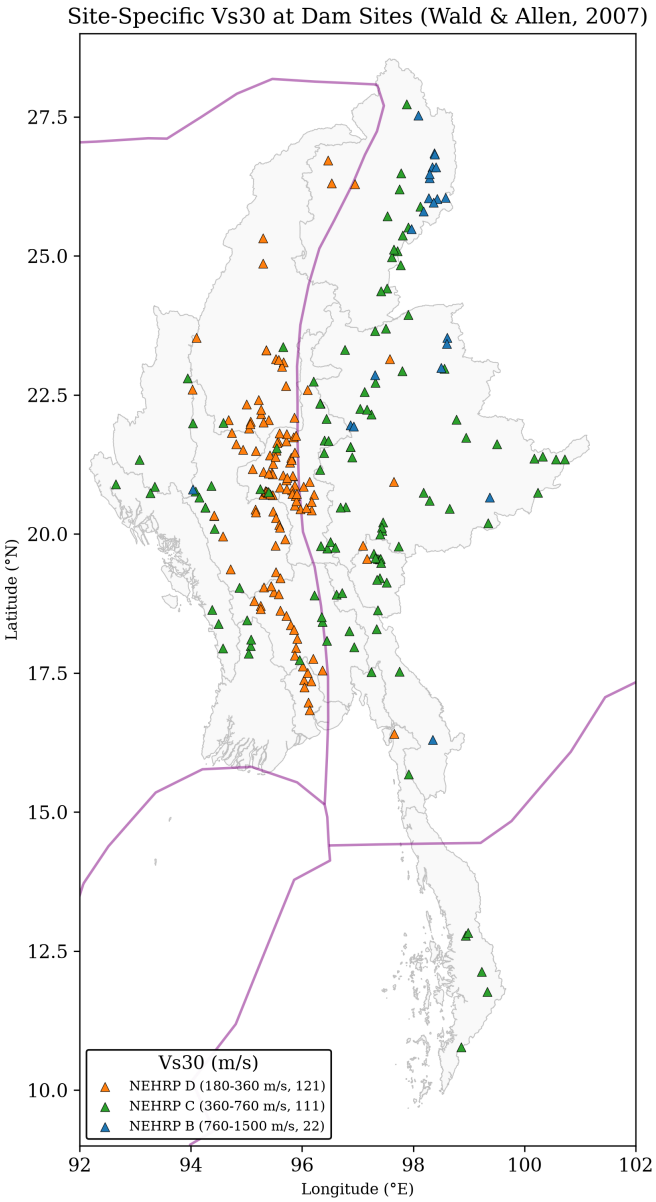


Figure 5: Site-specific V_{S30} at dam locations from topographic slope.

Table 2: Dam risk grades (corrected ASK08 GMPE).

Grade	Count	%
Critical	44	17.3%
High	116	45.7%
Moderate	45	17.7%
Low	49	19.3%

63.0% of the inventory is Critical or High under this heuristic screening scheme (thresholds 7, 5, and 3). The scores are not engineering safety classifications or calibrated failure probabilities. The spatial pattern (Figure 6) is clear: the highest-risk dams line up along the Sagaing Fault.

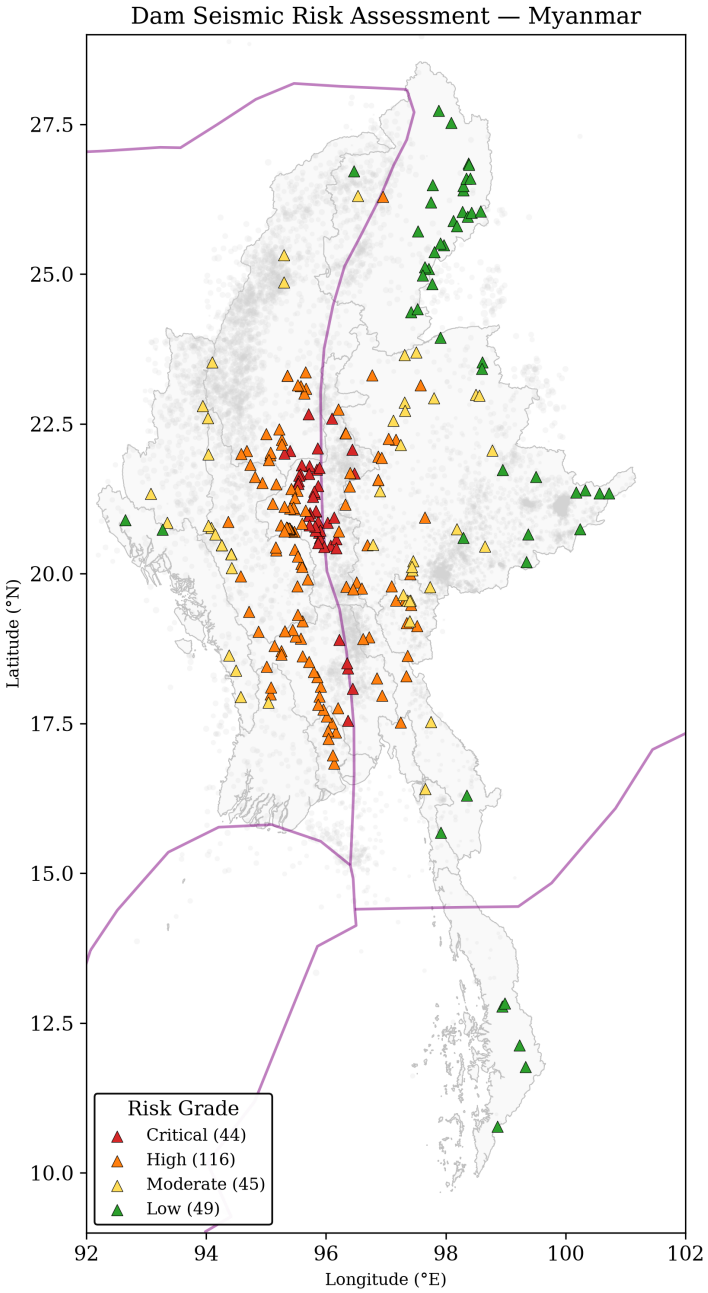


Figure 6: Dam risk map. The Sagaing Fault corridor dominates.

4.2 Sensitivity to Weights

The weights (0.35/0.30/0.20/0.15) are somewhat arbitrary, so I ran 100 Monte Carlo iterations with random Dirichlet-sampled weights. The median Critical+High fraction across draws is ∼67%, but the grading is genuinely weight-sensitive: 22 of the 100 draws fall below a 30% Critical+High fraction (minimum ∼12%, 5th percentile ∼17%), typically when the proximity weight dominates. The headline conclusion—that a substantial fraction of the portfolio carries elevated risk—holds for the large majority of weightings, but the exact grade counts should be read as conditional on the chosen weights.

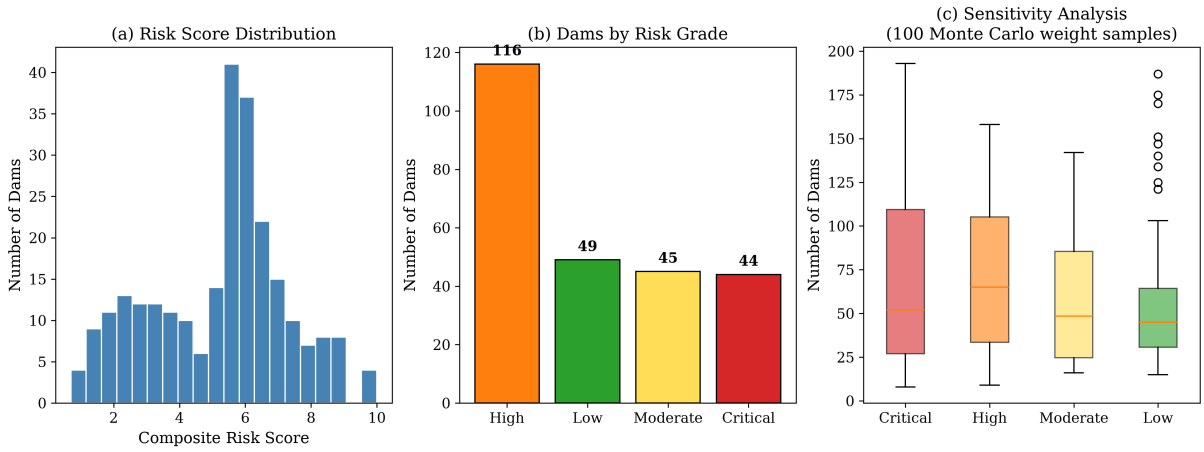


Figure 7: (a) Risk score histogram. (b) Grade counts. (c) Monte Carlo weight sensitivity.

5 Probabilistic Hazard

Beyond the single-event PGA, the Cornell-McGuire-style calculation integrates the G-R magnitude distribution and GMPE over the sampled M5–8 range, with fixed log-ground-motion variability ($\sigma_{ln} = 0.65$).

The 475-year PGA (approximately 10% chance of exceedance in 50 years under a stationary Poisson model) ranges from $0.002g$ to $0.562g$, with a mean of $0.143g$. The Frankel (1995)-style smoothed-seismicity source model uses the declustered catalog ($b = 1.04$) and integrates over source-to-site distance. 172 dams exceed $0.1g$ and 46 exceed $0.2g$. Annual exceedance *rate* λ is distinct from annual probability $1 - e^{-\lambda}$; the export now distinguishes these fields. These are *catalog-only estimates*: there is no fault-source recurrence term, and a single regional completeness model is applied despite network and temporal heterogeneity. AS08 is also extrapolated for distant source cells. Comparisons with Yang et al. (2023) and GEM are contextual only: their source models, locations, site conditions, and assumptions differ. Agreement in regional ranges does not validate this model or establish it as a lower bound.

6 Coulomb Stress Transfer

After a big earthquake, the stress field changes. Some areas get loaded (more likely to have aftershocks), others get unloaded ("stress shadow"). I modeled this using the point-source double-couple summation from Aki & Richards (2002).

The default calculation does *not* use the archived USGS 530-patch variable-slip solution. It constructs seven segments from an idealized Sagaing trace, divides them into patches, and assigns uniform slip from the M7.7 scalar moment (about 568 km total length and 1.56 m slip). This is a simplified point-source stress proxy, not a full elastic half-space solution. At ± 0.01 MPa, 16 dam locations are labeled positive ("Triggered") and 72 negative ("Shadow"). These describe the assumed crustal receiver geometry, not structural loading or a dam-failure probability. The four-lobed pattern and a symmetry test do not independently validate the stress amplitudes.

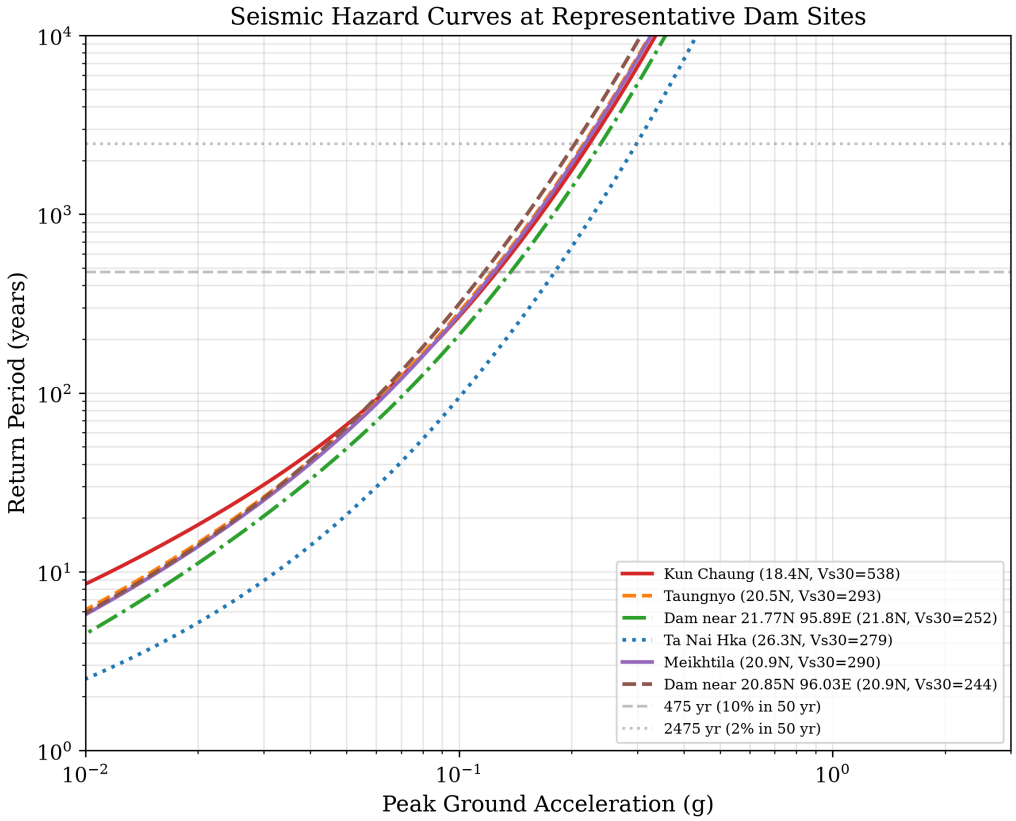


Figure 8: Hazard curves at representative dams. Dashed lines mark 475-year and 2,475-year return periods.

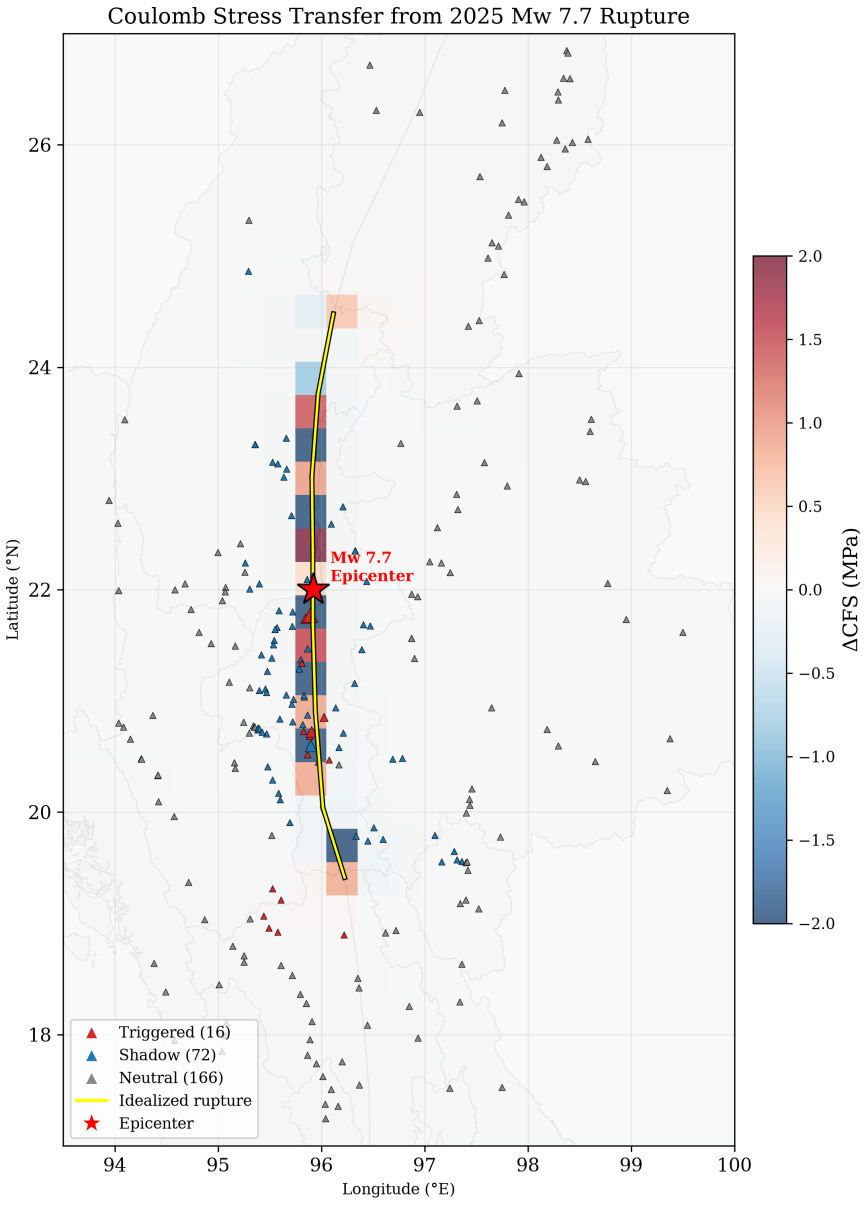


Figure 9: Simplified stress proxy from the idealized rupture. Marker colors and legend both use the exported ±0.01 MPa classification.

7 Fragility and Uncertainty

7.1 Dam Fragility Curves

The exploratory lognormal curves assign the largest state probability to "None" or "Slight" at most sites. Current outputs are in `docs/report/dam_fragility.json`. All dams are assumed earthfill because the inventory lacks construction types. Concrete and rockfill parameter sets are plotted for illustration only. These assumed medians are not calibrated Myanmar fragility curves or a direct implementation of HAZUS; they do not establish dam safety.

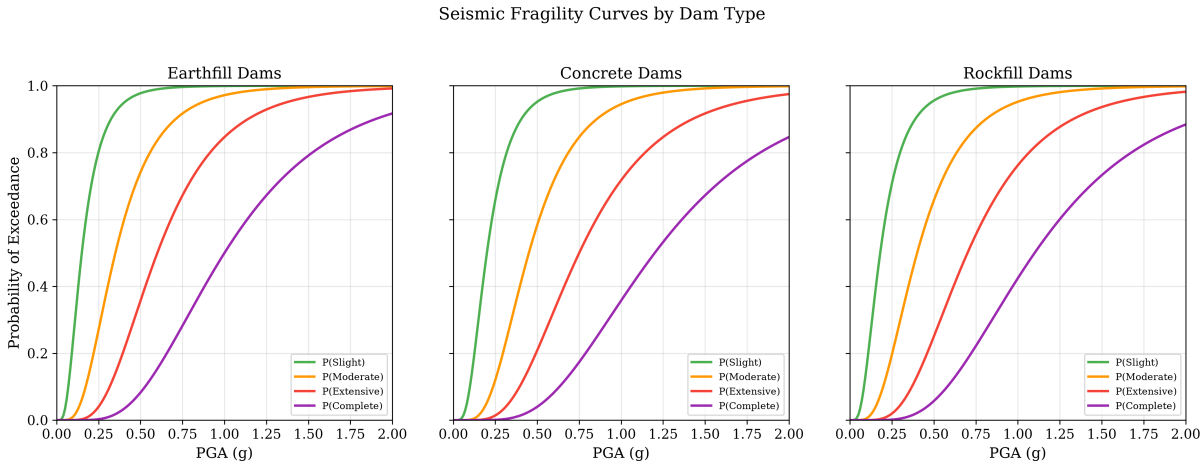


Figure 10: Fragility curves for three dam types.

7.2 Monte Carlo PGA Uncertainty

The GMPE has aleatory uncertainty ($\sigma_{ln} = 0.65$), meaning the actual PGA at any site could easily be 2× higher or lower than the median. The pipeline draws 1,000 lognormal samples per dam. These illustrate conditional ground-motion variability, not epistemic uncertainty in the source model, an annual event probability, or the probability of dam failure; current results are in `docs/report/monte_carlo_pga.csv`.

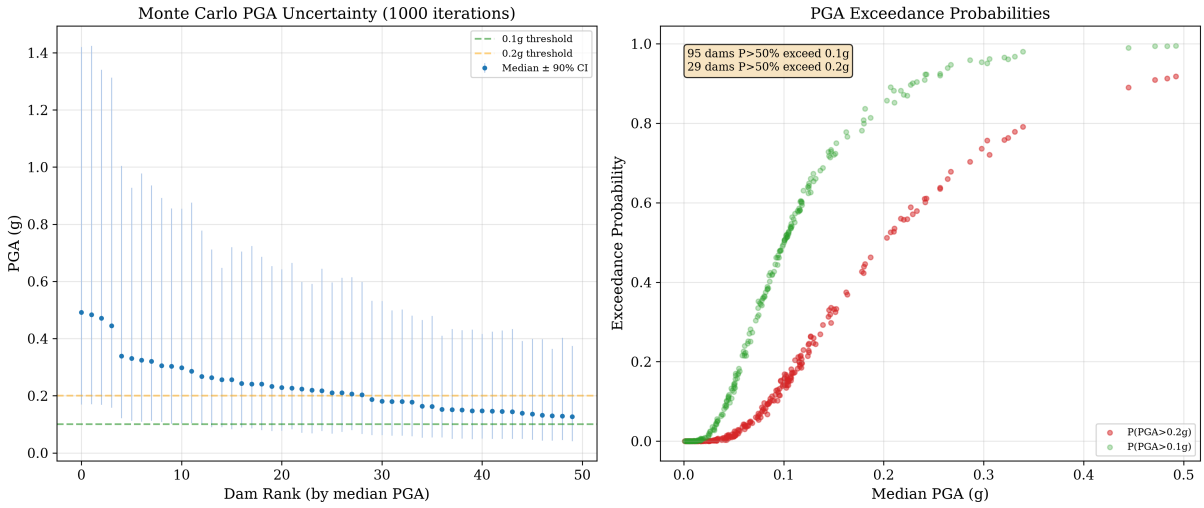


Figure 11: Monte Carlo PGA uncertainty. Error bars show 5th–95th percentile.

8 Population Exposure

The reproducible pipeline uses city-based population estimates, not a gridded population census. An earlier 2.9-million WorldPop claim came from an untracked offline raster analysis and is not independently reproducible from this repository; it is not retained as a verified result. Population within an epicentral radius is not a shaking- or inundation-based exposure estimate.

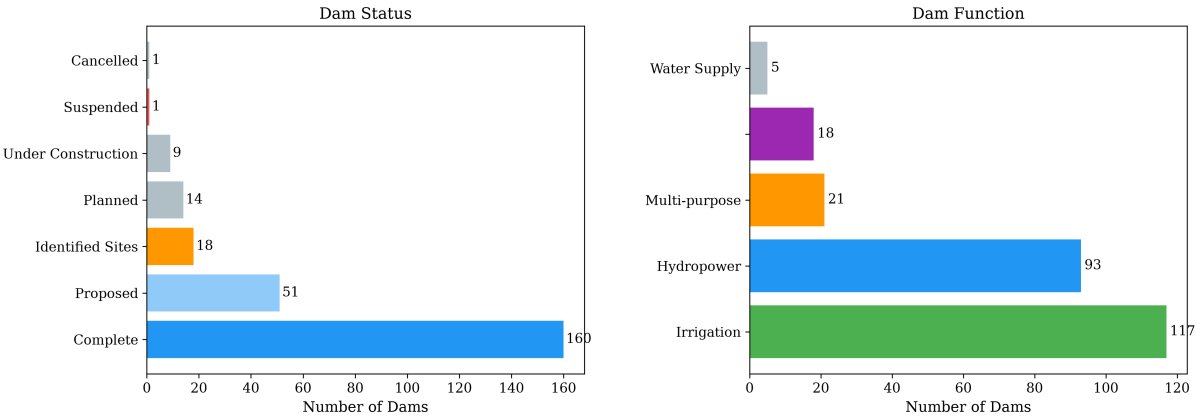


Figure 12: Dam inventory by status and function.

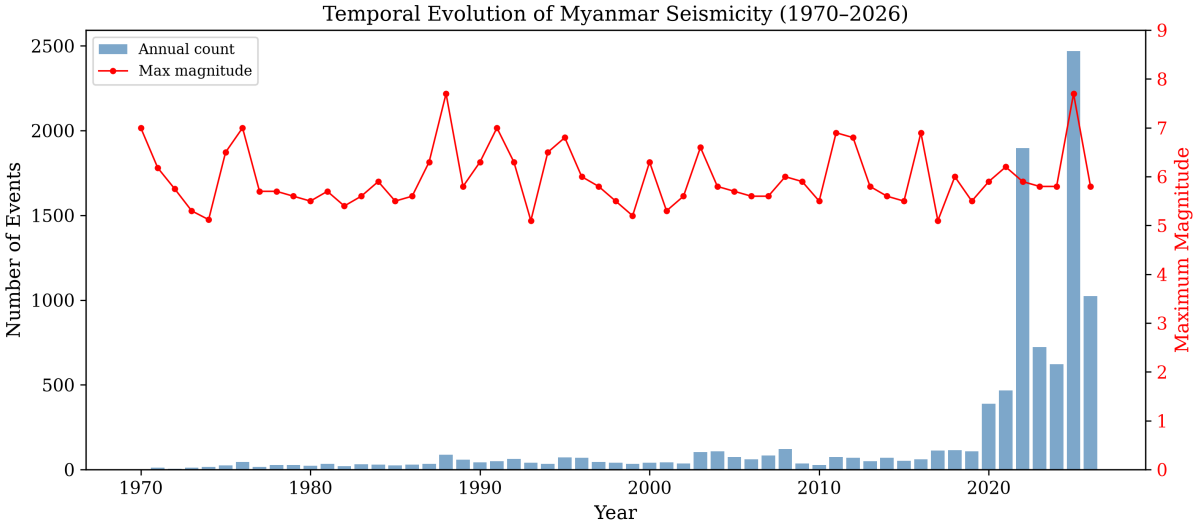


Figure 13: Annual catalog counts. Changes reflect both seismicity and changing network completeness; raw counts do not isolate either effect.

9 Building Exposure from OpenStreetMap

The OSM inventory contains 34,224 infrastructure elements, including schools, hospitals, and places of worship. These are mapped elements, not necessarily unique physical buildings or a complete census. For each element, the scenario uses distance to the actual 2025 rupture projection only, not to unrelated plate boundaries. The surface projection is an approximation to rupture-plane distance for the near-vertical fault. Modeled PGA-to-intensity bins place 3,395 elements in Severe (VIII) and 5,864 in Very Strong (VII); 1,674 schools and 583 hospitals have modeled PGA above 0.1g. These are scenario estimates, not observations of shaking or damage.

Using site-specific Vs30 from the USGS ShakeMap grid (mean 320 m/s, range 180–900 m/s)

instead of a flat 760 m/s reference rock makes a big difference—soft soils in the Irrawaddy basin amplify shaking significantly.

The repository lacks measured construction classes, occupancy, vulnerability, and damage observations sufficient for casualty estimation. The exploratory loss helper uses assumed coefficients and is not a verified HAZUS implementation; its outputs are not published here as injury, hospitalization, or collapse forecasts. Supershear/directivity effects are not modeled, but they do not justify attributing every residual to systematic underprediction.

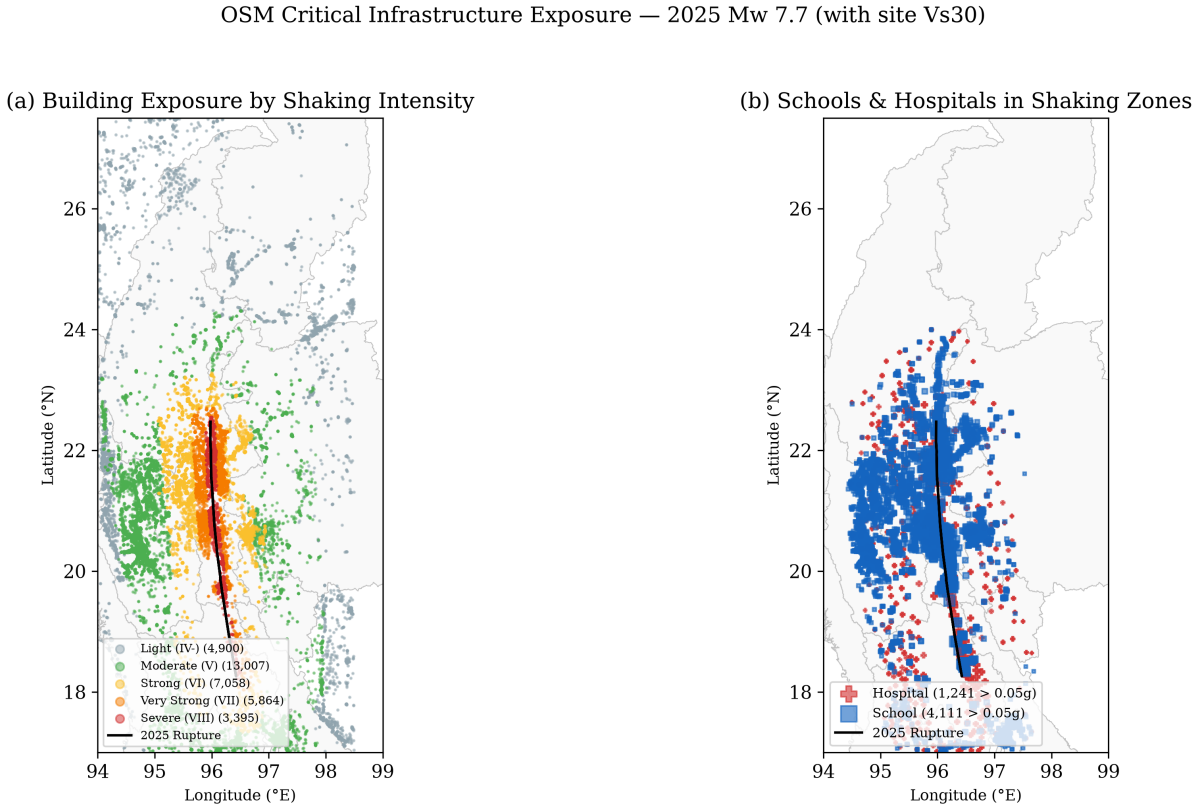


Figure 14: OSM scenario exposure. (a) 34,224 mapped elements colored by modeled shaking intensity. (b) Schools and hospitals sized by modeled PGA.

10 What I Learned

1. **Getting the GMPE right matters a lot.** The original implementation had wrong coefficients (a1 = −0.526 instead of 0.804) and no site response. Correct coefficients are necessary but not sufficient: component choice, rupture distance, quality flags, and site conditions must also match when checking observations.

2. **Catalog completeness is sneaky.** The raw b-value of 0.34 looked alarming but was just an artifact of mixing 1970s data (only M4.5+ detected) with 2020s data (M2+ detected) and the 2025 aftershock burst. A second caveat: the catalog pools magnitude types (~74% m_l, ~22% m_b, ~2% M_w) without converting to a common moment-magnitude scale, so the FMD slope carries some inter-scale bias on top of the completeness effect. Always check stability before trusting a b-value.

3. **Screening is not a safety assessment.** 160 inventory entries (63.0%) meet the chosen Critical/High thresholds. This inventory is not all Myanmar dams, and these heuristic grades do not demonstrate observed damage or structural failure risk.

4. **Open data makes this possible.** USGS ShakeMap, GEM faults, WorldPop, Copernicus DEM, Open Development Mekong—all free, all accessible via API or download. A hobby project can do real seismic hazard analysis.
5. **Validation is essential.** Without the ShakeMap station comparison, I wouldn't have caught the GMPE coefficient errors. A near-fault recording is useful, but one station cannot establish the validity of all downstream models.

10.1 Limitations

This is a hobby project, not a professional PSHA. Key caveats:

- No actual dam structural data (age, construction quality, foundation)
- Fragility curves use generic HAZUS parameters, not Myanmar-specific
- The AS08 model doesn't account for directivity (the supershear rupture would amplify PGA in the forward direction)
- Only 5 unflagged horizontal station pairs, with 3 within 200 km; geometric-mean peaks approximate GMRotI50
- Mixed magnitude types, nonuniform completeness, simplified source depths and rupture geometry, and uncalibrated screening weights
- The Coulomb model uses a simplified point-source approach, not the full Okada (1992) elastic half-space

11 How to Use the Code

Everything is at https://github.com/akzedevops/mmeqopendata.

```
git clone https://github.com/akzedevops/mmeqopendata.git
cd mmeqopendata
pip install -e ".[dev]"

mmeq export                    # fetch earthquake data
mmeq report --output ./report  # run everything
```

The `mmeq report` command generates maps, reports, dam scores, hazard curves, and exposure estimates. Run `python tools/build_figure_data.py` first for sensitivity inputs; `mmeq publish` synchronizes the paper tables, site downloads, and README with the generated report.

Live dashboard: https://akzedevops.github.io/mmeqopendata/

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